

TEST REPORT JIS C 62133-2:2020

Circular Notice of Interpretation for the Ministerial Order to provide of Technical Standards for Electrical Appliances and Materials Appendix 12, J62133-2(2021) (Lithium Ion Secondary)

Report Number.....	CMC250519008		
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Applicant's name.....	Xinxiang Hongli Supply Source Technology Co.,Ltd.		
Address.....	West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China		
Manufacturer's name.....	Xinxiang Hongli Supply Source Technology Co.,Ltd.		
Address.....	West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China		
Test specification:			
Standard.....	Interpretation for METI Ordinance of Technical Requirements (R07.01.31) Appendix 12, J 62133-2(2021), (JIS C 62133-2:2020): Lithium ion secondary batteries		
Test procedure.....	Type approved		
Non-standard test method.....	N/A		
Test result.....	Pass		
Test item description.....	Cylindrical Lithium-ion Rechargeable Cell		
Trade Mark.....	N/A		
Model/Type reference.....	IMR18650-1200mAh		
Ratings.....	3.7V, 1200mAh, 4.44Wh		
General disclaimer:			
The test results presented in this report relate only to the object tested. This report shall not be reproduced, except in full, without the written approval of the CMC. The authenticity of this Test Report and its contents can be verified by contacting the CMC, responsible for this Test Report.			

List of Attachments (including a total number of pages in each attachment):

Attachment 1: Photo documentation (On page 27)

Summary of testing:

Tests performed (name of test and test clause):

cl.7.2.1 Continuous charging at constant voltage (Secondary cells)
cl.7.2.2A Temperature cycling (Secondary cells)
cl.7.3.1 External short (Secondary cells)
cl.7.3.3 Free fall (Secondary cells)
cl.7.3.4 Thermal abuse (Secondary cells)
cl.7.3.5 Crush (Secondary cells)
cl.7.3.7 Forced discharge (Secondary cells)
cl.7.3.8A Low pressure (Secondary cells)
cl.7.3.8B High rate charge (Secondary cells)
cl.7.3.9 Forced internal short circuit (Secondary cells)

Testing location:

CMC Testing International (Shenzhen) Co., Ltd.
1-3/F., Building 7. He'er Hongben Industrial Zone,
Dawangshan Community, Shajing Subdistrict,
Baoan District, Shenzhen, Guangdong, China

Copy of marking plate:

The artwork below may be only a draft.

Cylindrical Lithium-ion Rechargeable Cell
IMR18650-1200mAh IMR19/66
+ 3.7V 1200mAh 4.44Wh -
Xinxiang Hongli Supply Source Technology Co.,Ltd.
Date Code: YYYYMMDD Made in China

Remark:

For the date code “YYYYMMDD”:

“YYYY” means the year of manufacture

“MM” means the month of manufacture

“DD” means the day of manufacture

Test item particulars..... :	
Classification of installation and use.....:	To be defined in final product
Supply connection.....:	Electrode plate
Recommend charging method declared by the manufacturer.....:	Charging the cell with 240mA constant current and 4.20V constant voltage until the current reduces to 24mA at ambient 20°C ± 5°C.
Discharge current (0,2 I_t A)	240mA
Specified final voltage..... :	3.0V
Upper limit charging voltage per cell.....:	4.25V
Maximum charging current..... :	600mA
Cell upper limit test temperature (T₃)..... :	45°C
Cell lower limit test temperature (T₂)..... :	0°C (test at -5°C)
Cell upper limit charging temperature (T₄).....:	45°C
Cell lower limit charging temperature (T₁)..... :	0°C (test at -5°C)
Charging temperature range of battery.....:	N/A
Polymer cell electrolyte type.....:	☐ gel polymer ☐ solid polymer ☒ N/A
Possible test case verdicts:	
- test case does not apply to the test object..... : N/A	
- test object does meet the requirement..... : P (Pass)	
- test object does not meet the requirement..... : F (Fail)	
Testing..... :	
Date of receipt of test item..... :	2025-05-19
Date (s) of performance of tests..... :	2025-05-19 to 2025-06-18
General remarks: The test results presented in this report relate only to the object tested. This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory. "(CXXX)" refers to sample number of batteries, "X" is 0~9; "(BXXX)" refers to sample number of batteries, "X" is 0~9; "(See Enclosure)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report.	
Throughout this report a ☐ comma / ☒ point is used as the decimal separator.	
Name and address of factory (ies)..... : Same as manufacturer.	

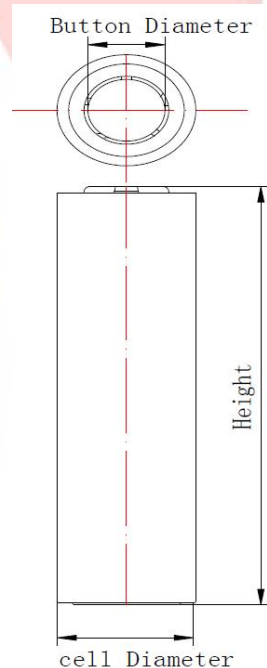
General product information:

The cell consists of the positive electrode plate, negative electrode plate, separator, electrolyte and case. The positive and negative electrode plates are housed in the case in the state being separated by the separator.

The main features of this model are shown as below:

Item	Cell
Model	IMR18650-1200mAh
Rated capacity	1200mAh
Nominal voltage	3.7V
Nominal charging current	240mA
Nominal discharging current	240mA
Maximum charging current	600mA
Maximum discharging current	600mA
Recommend charging Voltage	4.20V
Maximum Charge Voltage	4.25V
End-of-discharge Voltage	3.0V
Charge temperature range	0°C ~ 45°C
Recommend charging method declared by the manufacturer	Charging the cell with 240mA constant current until 4.20V, then constant voltage until charge current reduces to 24mA at ambient 20°C±5°C.

Construction:



cell Diameter: 18.5max Height: 65.5max
Cell (Unit: mm)

Circuit diagram:

None, cell only.

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)			
Clause	Requirement + Test	Result - Remark	Verdict
4	PARAMETER MEASUREMENT TOLERANCES		P
	Parameter measurement tolerances		P
5	GENERAL SAFETY CONSIDERATIONS		P
5.1	General		P
	Cells and batteries so designed and constructed that they are safe under conditions of both intended use and reasonably foreseeable misuse		P
5.2	Insulation and wiring	Cell only.	N/A
	The insulation resistance between the positive terminal and externally exposed metal surfaces of the battery (excluding electrical contact surfaces) is not less than 5 MΩ		N/A
	Insulation resistance (MΩ)..... :		—
	Internal wiring and insulation are sufficient to withstand maximum anticipated current, voltage and temperature requirements		N/A
	Orientation of wiring maintains adequate clearances and creepage distances between conductors		N/A
	Mechanical integrity of internal connections accommodates intended use		N/A
5.3	Venting		P
	Battery cases and cells incorporate a pressure relief mechanism or are constructed so that they relieve excessive internal pressure at a value and rate that will preclude rupture, explosion and self-ignition	Venting mechanism exists on top of the cylindrical cell.	P
	Encapsulation used to support cells within an outer casing does not cause the battery to overheat during normal operation nor inhibit pressure relief		N/A
5.4	Temperature, voltage and current management	Cell only.	N/A
	Secondary batteries are designed such that abnormal temperature rise conditions are prevented		N/A
	Secondary batteries are designed to be within temperature, voltage and current limits specified by the cell manufacturer		N/A
	The cell manufacturer are provided the information of temperature, voltage and current limits to the battery manufacturer		N/A
	The battery manufacturer are provided the information of temperature, voltage and current limits to the equipment manufacturer		N/A
5.5	Terminal contacts		P

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
	The size and shape of the terminal contacts ensure that they can carry the maximum anticipated current	Electrode plate contacts complied with the requirements.	P
	External terminal contact surfaces are formed from conductive materials with good mechanical strength and corrosion resistance	Complied.	P
	Terminal contacts are arranged to minimize the risk of short circuits		P
5.6	Assembly of cells into batteries		N/A
5.6.1	General	Cell only.	N/A
	Each secondary battery have an independent control and protection for current, voltage, temperature and any other parameter required for safety and to maintain the cells within their operating region		N/A
	This protection may be provided external to the battery such as within the charger or the end devices		N/A
	If protection is external to the secondary battery, the manufacturer of the battery provide this safety relevant information to the external device manufacturer for implementation		N/A
	If there is more than one secondary battery housed in a single battery case, each secondary battery have protective circuitry that can maintain the cells within their operating regions		N/A
	Manufacturers of cells specify current, voltage and temperature limits so that the battery manufacturer/designer may ensure proper design and assembly		N/A
	Secondary batteries that are designed for the selective discharge of a portion of their series connected cells incorporate circuitry to prevent operation of cells outside the limits specified by the cell manufacturer		N/A
	Protective circuit components are added as appropriate and consideration given to the end-device application		N/A
	The manufacturer of the battery provide a safety analysis of the battery safety circuitry with a test report including a fault analysis of the protection circuit under both charging and discharging conditions confirming the compliance		N/A
5.6.2	Design recommendation	Cell only.	N/A

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
	For the secondary battery consisting of a single secondary cell or a single cellblock, it is recommended that the charging voltage of the cell does not exceed the upper limit of the charging voltage specified in Table 2		N/A
	For the secondary battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that the voltages of any one of the single cells or single cellblocks does not exceed the upper limit of the charging voltage, specified in Table 2, by monitoring the voltage of every single cell or the single cellblocks		N/A
	For the secondary battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that charging is stopped when the upper limit of the charging voltage is exceeded for any one of the single cells or single cellblocks by measuring the voltage of every single cell or the single cellblocks		N/A
	For secondary batteries consisting of series-connected cells or cell blocks, nominal charge voltage are not counted as an overcharge protection		N/A
	For secondary batteries consisting of series-connected cells or cell blocks, cells have closely matched capacities, be of the same design, be of the same chemistry and be from the same manufacturer		N/A
	It is recommended that the secondary cells and cell blocks are not discharged beyond the cell manufacturer's specified final voltage		N/A
	For secondary batteries consisting of series-connected secondary cells or cell blocks, cell balancing circuitry are incorporated into the battery management system		N/A
5.6.3	Mechanical protection for secondary cells and components of secondary batteries	Cell only.	N/A
	Mechanical protection for secondary cells, cell connections and control circuits within the secondary battery are provided to prevent damage as a result of intended use and reasonably foreseeable misuse		N/A
	The mechanical protection can be provided by the battery case or it can be provided by the end product enclosure for those batteries intended for building into an end product		N/A

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
	The battery case and compartments housing cells are designed to accommodate cell dimensional tolerances during charging and discharging as recommended by the cell manufacturer		N/A
	For secondary batteries intended for building into a portable end product, testing with the battery installed within the end product is considered when conducting mechanical tests		N/A
5.6.3A	Prevention for sharp corner hazard	Cell only.	N/A
	Except in the case of necessary function, concavo-convex or sharp corner not exist to cause hazards for cell or battery		N/A
	When the corner exist on the cell or battery enclosure or connection parts, mechanical protection provided to prevent user contact		N/A
	For cell or battery not intended to be handled by end user, the protection applied can be decided by agreement between the cell manufacturer and the battery and/or end product manufacturer, Conformity is checked by inspection		N/A
5.7	Quality plan		P
	The cell and battery manufacturer prepares and implements a quality plan that defines procedures for the inspection of materials, components, secondary cells and batteries and which covers the whole process of producing each type of secondary cell or battery. The cell and battery manufacturers should understand their process capabilities and should institute the necessary process controls as they relate to product safety	Complied. ISO 9001:2015 certificate provided.	P
5.8	Battery safety components	Cell only.	N/A
6	TYPE TEST AND SAMPLE SIZE		P
	Tests are made with the number of secondary cells or batteries specified in Table 1 using secondary cells or batteries that are not more than six months old		P
	The internal resistance of coin cells are measured in accordance with Annex D. Coin cells with internal resistance less than or equal to 3 Ω are tested in accordance with Table 1. No test is required for coin cells with internal resistance greater than 3 Ω	Not coin cells.	N/A
	Unless otherwise specified, tests are carried out in an ambient temperature of 20 $^{\circ}\text{C} \pm 5^{\circ}\text{C}$		P

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
	The safety analysis of 5.6.1 identify those components of the protection circuit that are critical for short-circuit, overcharge and over discharge protection	Cell only.	N/A
	When conducting the short-circuit test, consideration is given to the simulation of any single fault condition that is likely to occur in the protecting circuit that would affect the short-circuit test	Cell only.	N/A

7	SPECIFIC REQUIREMENTS AND TESTS		P
7.0A	General		P
7.1	Charging procedure for test purposes		P
7.1.0A	The first and second procedures are specified as charging procedures for tests. These charging procedures, however, do not apply to 7.3.6, 7.3.7, 7.3.8B and 7.3.8D where the charging process is the purpose of the test		P
7.1.1	First procedure		P
	Unless otherwise specified in this Standard, the secondary cells and batteries shall be charged in an ambient temperature of 20 °C ± 5 °C, using the method declared by the manufacturer	See page 4.	P
	Prior to charging, the secondary cell and battery shall have been discharged at an ambient temperature of 20 °C ± 5 °C at a constant current of 0.2 It A down to the designed final voltage specified by the manufacturer	See page 4.	P
	This charging procedure applies to 7.2.1, 7.2.2, 7.2.2A, 7.3.2, 7.3.3, 7.3.8.1, 7.3.8.2, 7.3.8A and 7.3.8C		P
7.1.2	Second procedure		P
	After stabilization for 1 h and 4 h, respectively, at ambient temperature of upper limit test temperature and lower limit test temperature, as specified in Table 2, cells are charged by using the upper limit charging voltage and maximum charging current, until the charging current at the time of constant voltage charge control is reduced to 0.05 It A	Charge temperature specified by manufacturer: 0-45°C. -5°C used for lower limit tests. 45°C used for upper limit tests.	P
	This charging procedure applies only to 7.3.1, 7.3.4, 7.3.5, and 7.3.9		P
7.2	Intended use		P
7.2.1	Continuous charging at constant voltage (secondary cells)	Tested complied.	P

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Clause	Requirement + Test	Result - Remark	Verdict
	Fully charged secondary cells, according to the first procedure in 7.1.1, are subjected for 28 days to a charge at upper limit charging voltage and upper limit test temperature. After the test, visual inspection shall be performed	Charging for 28 days with 4.25V at 45°C.	P
	Results: No fire. No explosion. No leakage..... :	(See appended table 7.2.1).	P
7.2.2	Moulded case stress at high ambient temperature (Secondary batteries)	Cell only.	N/A
	Fully charged secondary batteries, according to the first procedure in 7.1.1, are exposed to a high temperature. The secondary battery is placed in an air circulating oven at a temperature of 70 °C ± 2 °C. The batteries remain in the oven for 7 h, after which they are removed and allowed to return to an ambient temperature of 20 °C ± 5 °C		N/A
	Oven temperature (°C)..... :		—
	Results: No physical distortion of the battery casing resulting in exposure of internal components		N/A
7.2.2A	Temperature cycle (secondary cells)	Tested complied.	P
	Fully charged secondary cells or batteries, according to the first procedure in 7.1.1, are subjected to temperature cycling (–20 °C, +75 °C), in forced draught chambers, according to the following procedure and the temperature profile shown in Figure 0A Step 1: Place the secondary cells or batteries in an temperature of 75 °C ± 2 °C for 4 h Step 2: Change the temperature to 20 °C ± 5 °C within 30 min and maintain at this temperature for not less than 2 h Step 3: Change the temperature to –20 °C ± 2 °C within 30 min and maintain at this temperature for 4 h Step 4: Change the temperature to 20 °C ± 5 °C within 30 min and maintain at this temperature for not less than 2 h Step 5: Repeat Steps 1 to 4 for a further four cycles. Transition from Step 4 to Step 1 within 30 min Step 6: After the fifth cycle, store the secondary cells or batteries at 20 °C ± 5 °C for 7 days and then conduct a visual inspection		P
	Results: No fire. No explosion. No leakage	No fire. No explosion. No leakage.	P
7.3	Reasonably foreseeable misuse		P

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)			
Clause	Requirement + Test	Result - Remark	Verdict
7.3.1	External short-circuit (secondary cells)	Tested complied.	P
	Charging procedure: 7.1.2		P
	Ambient temperature: 55 °C ± 5 °C	(See appended table 7.3.1).	P
	Resistance of circuit (mΩ): 80 mΩ ± 20 mΩ	(See appended table 7.3.1).	P
	The cells were tested until one of the following occurred:		P
	- 24 hours elapsed; or		N/A
	- The case temperature declined by 80% of the maximum temperature rise		P
	Results: No fire. No explosion..... :	(See appended table 7.3.1).	P
7.3.2	External short-circuit (secondary batteries)	Cell only.	N/A
	Charging procedure: 7.1.1		N/A
	Ambient temperature: 20 °C ± 5 °C		N/A
	Resistance of circuit (mΩ): 80 mΩ ± 20 mΩ		N/A
	The batteries were tested until one of the following occurred:		N/A
	- 24 hours elapsed; or		N/A
	- The case temperature declined by 80 % of the maximum temperature rise		N/A
	In case of rapid decline in short circuit current, the battery pack remained on test for an additional one hour after the current reached a low end steady state condition		N/A
	A single fault in the discharge protection circuit is conducted on one to four (depending upon the protection circuit) of the five samples before conducting the short-circuit test		N/A
	A single fault applies to protective component parts such as MOSFET (metal oxide semiconductor field-effect transistor), fuse, thermostat or positive temperature coefficient (PTC) thermistor		N/A
	Results: No fire. No explosion..... :		N/A
7.3.3	Free fall (secondary cells)	Tested complied.	P
	Fully charged secondary cells or batteries, according to the first procedure in 7.1.1, are stored in an ambient temperature of 20 °C ± 5 °C. Each secondary cell or battery is dropped three times from a height of 1.0 m ± 0.01 m onto a flat concrete floor or metal floor. The cells or batteries are dropped so as to obtain impacts in random orientations. After the test, the secondary cell or battery shall be put on rest for a minimum of 1 h and then a visual inspection shall be performed		P

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)			
Clause	Requirement + Test	Result - Remark	Verdict
	Results: No fire. No explosion	No fire. No explosion.	P
7.3.4	Thermal abuse (secondary cells)	Tested complied.	P
	Oven temperature (°C).....: 130.2		—
	Results: No fire. No explosion.	No fire. No explosion.	P
7.3.5	Crush (secondary cells)	Tested complied.	P
	The crushing force was released upon:		P
	- The maximum force of 13 KN ± 0.78 KN has been applied; or		P
	- An abrupt voltage drop of one-third of the original voltage has been obtained		N/A
	A cylindrical or prismatic secondary lithium cell is crushed with its longitudinal axis parallel to the flat surfaces of the crushing apparatus. Test only the wide side of prismatic secondary lithium cells. A coin cell shall be crushed by applying the force on its flat surface	Cylindrical cell.	P
	Results: No fire. No explosion..... :	(See appended table 7.3.5).	P
7.3.6	Overcharge (secondary batteries)	Cell only.	N/A
	The test shall be carried out in an ambient temperature of 20 °C ± 5 °C. Each secondary battery shall be discharged at a constant current of 0.2 It A, to the designed final voltage specified by the manufacturer		N/A
	The supply voltage which is:		N/A
	- 1.4 times the upper limit charging voltage presented in Table A.1 (but not to exceed 6,0 V) for single cell/cell block batteries or		N/A
	- 1.2 times the upper limit charging voltage resented in Table A.1 per cell for series connected multi-cell batteries, and		N/A
	- Sufficient to maintain a current of 2,0 It A throughout the duration of the test or until the supply voltage is reached		N/A
	Test was continued until the temperature of the outer casing		N/A
	- Reached steady state conditions (less than 10 °C change in 30-minute period); or		N/A
	- Returned to ambient		N/A
	Results: No fire. No explosion..... :		N/A
7.3.7	Forced discharge (secondary cells)	Tested complied.	P
	A secondary cell is discharged to the designed final voltage specified by the cell manufacturer	Lower limit discharge voltage 3.0V.	P

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Clause	Requirement + Test	Result - Remark	Verdict
	The discharged cell is then subjected to a reverse charge at 1 It A for 90 min		P
	- The discharge voltage reaches the negative value of upper limit charging voltage within the testing duration. The voltage is maintained at the negative value of the upper limit charging voltage by reducing the current for the remainder of the testing duration		N/A
	- The discharge voltage does not reach the negative value of upper limit charging voltage within the testing duration. The test is terminated at the end of the testing duration		P
	Results: No fire. No explosion..... :	(See appended table 7.3.7).	P
7.3.8	Mechanical tests (secondary batteries)		N/A
7.3.8.1	Vibration (secondary batteries)	Cell only.	N/A
	Results: no fire, no explosion or no leakage..... :		N/A
7.3.8.2	Mechanical shock (secondary batteries)	Cell only.	N/A
	Results: No fire, no explosion or no leakage..... :		N/A
7.3.8A	Low pressure (secondary cells)	Tested complied.	P
	Charging procedure: 7.1.1		P
	Ambient temperature: 20 °C ± 5 °C		P
	Air pressure: equal to or less than 11.6 kPa (simulates an altitude of 15 240 m)		P
	Duration: 6 h		P
	Results: No fire, no explosion or no leakage..... :	No fire. No explosion. No leakage.	P
7.3.8B	High-rate charge (secondary cells)	Tested complied.	P
	The test shall be performed at upper limit test temperature or lower limit test temperature		P
	When the discharged secondary cell is fully charged with the charging current of three times the maximum charging current or when the corresponding electronic device or secondary battery before full charge has a protection element, charging shall be performed until charging current is broken by the actuation of the safety device of the protection element	Charged until the cell is fully charged.	P
	Results: No fire, no explosion..... :	No fire, No explosion.	P
7.3.8C	Falling of secondary battery installed in a device (secondary batteries)	Cell only.	N/A
	Battery equipped with Device is tested		N/A

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
	The battery that is charged according to 7.1.1 is installed in the portable electronic application to be used, or subjected to the condition, simulating the actual use Then, it is dropped once in the direction most likely to affect in a negative manner from the height, which is specified in JIS C 6950 or JIS C6065, according to the portable electronic applications, in which there batteries are assumed to be installed, on to a concrete floor. An iron plate may be used in place of the concrete floor		N/A
	Requirement: External short circuit shall not be caused inside of the battery, and internal short circuit shall not be caused in cells contained in the battery.		N/A
7.3.8D	Overcharge protection (secondary batteries)	Cell only.	N/A
	Ambient temperature: 20 °C ± 5 °C		N/A
	One of the following test is conducted		N/A
	a) When the secondary battery consists of secondary cells or one step of cell block, measure the voltage applied to the secondary cells or the cell block at the time of charge		N/A
	b) In the case of the structure of the secondary battery in which two or more secondary cells or cell blocks are connected in series, charge while measuring the voltage of each secondary cell or cell block. At the same time, discharge single secondary cell or cell block gradually and compulsorily, and measure the voltage of each secondary cell or cell block of others		N/A
	c) In the case of the structure of the secondary battery in which two or more secondary cells or cell blocks are connected in series, apply the voltage which exceeds the upper limit charging voltage in Table 2 to the secondary cell or cell block while measuring the voltage of each secondary cell or cell block, and measure the voltage when charge stops		N/A
	The battery provides with protective circuits		N/A
	Appliance in which battery is installed or battery charger provides with protective circuits		N/A
	Requirement: cells or cellblocks shall not exceed upper limit charging voltage		N/A
7.3.9	Forced internal short-circuit (secondary cells)	Tested complied.	P
	The pressing was stopped upon:		P
	- A voltage drop of 50 mV has been detected; or		N/A

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
	- The pressing force of 800 N (cylindrical cells) or 400 N (prismatic cells) has been reached	800N for cylindrical cell.	P
	Results: No fire..... :	(See appended table 7.3.9).	P

8	INFORMATION FOR SAFETY		P
8.1	General		P
	Manufacturers of secondary cells provides information about current, voltage and temperature limits of their products	Information for safety mentioned in manufacturer's specifications.	P
	Manufacturers of batteries provides information regarding how to minimize and mitigate hazards to equipment manufacturers or end-users	Cell only.	N/A
	Systems analyses are performed by device manufacturers to ensure that a particular battery design prevents hazards from occurring during use of a product		N/A
	As appropriate, any information relating to hazard avoidance resulting from a system analysis is provided to the end user		N/A
8.2	Small cell and battery safety information	Not small cells.	N/A
	The following warning language is to be provided with the information packaged with the small cells and batteries and equipment using them		N/A
	- Keep small cells and batteries which are considered swallowable out of the reach of children		N/A
	- Swallowing may lead to burns, perforation of soft tissue, and death. Severe burns can occur within 2 h of ingestion		N/A
	- In case of ingestion of a cell or battery, seek medical assistance promptly		N/A

9	MARKING		P
9.1	Secondary cell marking		P
	Secondary cells are marked as specified in JIS C 8711, except coin cells	The cell is marked in accordance with JIS C 8711, also see copy of marking plate.	P
	Coin cells whose external surface area is too small to accommodate the markings on the cells show the designation and polarity	Not coin cells.	N/A
	By agreement between the secondary cell manufacturer and the battery and/or end product manufacturer, component secondary cells used in the manufacture of a battery need not be marked		N/A

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)			
Clause	Requirement + Test	Result - Remark	Verdict
	However, when component secondary cells are not marked, the secondary cell marking are indicated with the secondary battery, the instructions or the specifications		N/A
9.2	Secondary battery marking	Cell only.	N/A
	Secondary batteries are marked as specified in JIS C 8711, except for coin batteries		N/A
	Coin batteries whose external surface area is too small to accommodate the markings on the batteries show the designation and polarity		N/A
	Secondary batteries are marked with an appropriate caution statement		N/A
	- Terminals have clear polarity marking on the external surface of the battery, or		N/A
	- Not be marked with polarity markings if the design of the external connector prevents reverse polarity connections		N/A
9.3	Caution for ingestion of small secondary cells and batteries	Not small cells.	N/A
	Small secondary cells and batteries determined to be small according to 8.2 are included a caution statement regarding the hazards of ingestion in accordance with 8.2		N/A
	Small secondary cells and batteries are intended for direct sale in consumer-replaceable applications, caution for ingestion is given on the immediate package	Not intended for direct sale.	N/A
9.4	Other information	Cell only.	N/A
	The following information are marked on the secondary battery or supplied to the equipment manufacturer by the battery manufacturer		N/A
	- Storage and disposal instructions		N/A
	- Recommended charging instructions		N/A
10	PACKAGING AND TRANSPORT		N/A
	Packaging for coin cells and small secondary batteries are not be small enough to fit within the limits of the ingestion gauge of Figure 3	Not coin cells.	N/A

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
ANNEX A	CHARGING AND DISCHARGING RANGE OF SECONDARY LITHIUM ION CELLS FOR SAFE USE		P
A.1	General		P
A.2	Safety of lithium ion secondary battery	Complied.	P
A.3	Consideration on charging voltage	Complied.	P
A.3.1	General		P
A.3.2	Upper limit charging voltage	4.25V applied.	P
A.3.2.1	General		P
A.3.2.2	Explanation of safety viewpoint		P
A.3.2.3	Safety requirements, when different upper limit charging voltage is applied	4.25V applied.	N/A
A.4	Consideration of temperature and charging current		P
A.4.1	General		P
A.4.2	Recommended temperature range		P
A.4.2.1	General		P
A.4.2.2	Safety consideration when a different recommended temperature range is applied	Recommended temperature range declared by client is: 0°C to 45°C.	P
A.4.3	High temperature range	Not higher than the temperature range specific in this standard.	N/A
A.4.3.1	General		N/A
A.4.3.2	Explanation of safety viewpoint		N/A
A.4.3.3	Safety considerations when specifying charging conditions in the high temperature range		N/A
A.4.3.4	Safety considerations when specifying a new upper limit in the high temperature range		N/A
A.4.4	Low temperature range	Low temperature declared by client is: 0°C.	P
A.4.4.1	General		P
A.4.4.2	Explanation of safety viewpoint		P
A.4.4.3	Safety considerations, when specifying charging conditions in the low temperature range		P
A.4.4.4	Safety considerations when specifying a new lower limit in the low temperature range	-5°C applied for testing in this report for safety considerations.	P
A.4.5	Scope of the application of charging current		P
A.4.5.A	Determination of model adoption		P

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)			
Clause	Requirement + Test	Result - Remark	Verdict
A.4.6	Consideration of discharge		P
A.4.6.1	General		P
A.4.6.2	Final voltage and explanation of safety viewpoint	Cell specified final voltage 3.0V, not exceed 3.0V specified by cell manufacturer.	P
A.4.6.3	Discharge current and temperature range		P
A.4.6.4	Scope of application of the discharging current		P
A.5	Sample preparation		P
A.5.1	General		P
A.5.2	Insertion procedure for nickel particle to generate internal short-circuit		P
A.5.3	Disassembly of charged cell		P
A.5.4	Shape and material of nickel particle		P
A.5.5	Insertion of nickel particle in cylindrical lithium ion secondary cell		P
A.5.5.1	Insertion of nickel particle in winding core		P
A.5.5.2	Marking the position of the nickel particle on both ends of the winding core of the separator		P
A.5.6	Insertion of nickel particle in prismatic lithium ion secondary cell		N/A
A.6	Experimental procedure of the forced internal short-circuit test		P
A.6.1	Material and tools for preparation of nickel particle		P
A.6.2	Example of a nickel particle preparation procedure		P
A.6.3	Positioning (or placement) of a nickel particle		P
A.6.4	Damaged separator precaution		P
A.6.5	Caution for rewinding separator and electrode		P
A.6.6	Insulation film for preventing short-circuit		P
A.6.7	Caution when disassembling a cell		P
A.6.8	Protective equipment for safety		P
A.6.9	Caution in the case of fire during disassembling		P
A.6.10	Caution for the disassembling process and pressing the electrode core		P
A.6.11	Recommended specifications for the pressing device		P

Interpretation for METI Ordinance of Technical Requirements, Appendix 12, J62133-2 (2021) (JISC 62133-2:2020)

Clause	Requirement + Test	Result - Remark	Verdict
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ANNEX B	RECOMMENDATIONS TO EQUIPMENT MANUFACTURERS AND BATTERY ASSEMBLERS		N/A
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ANNEX C	RECOMMENDATIONS TO THE END-USERS		N/A
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ANNEX D	MEASUREMENT OF THE INTERNAL AC RESISTANCE FOR COIN CELLS		N/A
D.1	General		N/A
D.2	Method		N/A
	A sample size of three coin cells is required for this measurement		N/A
	Coin cells with an internal resistance greater than 3 Ω require no further testing.....:		N/A
	Coin cells with an internal resistance less than or equal to 3 Ω are subjected to the testing according to Clause 6 and Table 1		N/A

ANNEX E	PACKAGING AND TRANSPORT		N/A
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ANNEX F	COMPONENT STANDARDS REFERENCES		N/A
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TABLE: Critical components information					P
Object / part No.	Manufacturer / trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾
Positive electrode	Xinxiang hongli Power Science and Technology Co., Ltd.	HL04	LiMn ₂ O ₄	--	--
Negative electrode	Henan Zhonglian Gaoke new energy Co., Ltd	ZL138	C	--	--
Electrolyte	Xinxiang Huarui Lithium New Energy Co., Ltd	HR-HL01	LiPF ₆	--	--
Separator	Henan Huiqiang New Energy Materials Technology Co., Ltd	25um	PP, 0.025mm*61mm, Shutdown temperature: 150°C	--	--
Supplementary information:					

7.2.1	TABLE: Continuous charging at constant voltage (Secondary cells)				P
Sample No.	Upper limit charging voltage V_c , (Vdc)	Upper limit test temperature, ($^{\circ}\text{C}$)	OCV at start of test, (Vdc)	Results	
SN250519008C001	4.25	45	4.20	P	
SN250519008C002	4.25	45	4.20	P	
SN250519008C003	4.25	45	4.20	P	
SN250519008C004	4.25	45	4.20	P	
SN250519008C005	4.25	45	4.20	P	
Supplementary information: - No fire or explosion - No leakage					

7.3.1	TABLE: External short circuit (Secondary cells)					P
Sample No.	Ambient, ($^{\circ}\text{C}$)	OCV at start of test, (Vdc)	Resistance of circuit, ($\text{m}\Omega$)	Maximum case temperature rise ΔT ($^{\circ}\text{C}$)	Results	
Samples charged at charging temperature upper limit (45°C)						
SN250519008C011	56.3	4.22	89	119.4	P	
SN250519008C012	56.3	4.21	88	117.2	P	
SN250519008C013	56.3	4.21	92	115.2	P	
SN250519008C014	56.3	4.22	94	120.5	P	
SN250519008C015	56.3	4.22	86	118.8	P	
Samples charged at charging temperature lower limit (-5°C)						
SN250519008C016	56.1	4.18	82	117.6	P	
SN250519008C017	56.1	4.17	89	118.8	P	
SN250519008C018	56.1	4.17	85	115.7	P	
SN250519008C019	56.1	4.18	83	119.5	P	
SN250519008C020	56.1	4.18	90	113.6	P	
Supplementary information: - No fire or explosion						

7.3.2	TABLE: External short circuit (Secondary batteries)					N/A
Sample No.	Ambient T (°C)	OCV before test (Vdc)	Resistance of circuit (mΩ)	Maximum case temperature rise ΔT (°C)	Component single fault condition	Results
Supplementary information:						

7.3.5	TABLE: Crush (Secondary cells)				P
Sample No.	OCV before test (Vdc)	OCV at removal of crushing force (Vdc)	Maximum force applied to the cell during crush (KN)	Results	
Samples charged at charging temperature upper limit (45°C)					
SN250519008C034	4.21	4.21	13.03	P	
SN250519008C035	4.22	4.21	13.02	P	
SN250519008C036	4.22	4.22	13.09	P	
SN250519008C037	4.21	4.21	13.05	P	
SN250519008C038	4.21	4.21	13.04	P	
Samples charged at charging temperature lower limit (-5°C)					
SN250519008C039	4.18	4.18	13.01	P	
SN250519008C040	4.17	4.17	13.04	P	
SN250519008C041	4.18	4.18	13.07	P	
SN250519008C042	4.18	4.17	13.10	P	
SN250519008C043	4.18	4.17	13.06	P	
Supplementary information:					
- No fire or explosion					
Note: - A 13KN force applied at the longitudinal axis of cylindrical cells. No voltage abrupt drop occurred.					



7.3.6	TABLE: Overcharge (Secondary batteries)				N/A
Constant charging current (A).....:					—
Supply voltage (Vdc).....:					—
Sample No.	OCV before charging, (Vdc)	Total charging time (minute)	Maximum outer casing temperature, (°C)	Results	
Supplementary information:					

7.3.7	TABLE: Forced discharge (Secondary cells)				P
Sample No.	OCV before application of reverse charge (Vdc)	Measured reverse charge I _t (A)	End-of-discharge voltage (Vdc)	Results	
SN250519008C044	3.09	1.2	3.0	P	
SN250519008C045	3.08	1.2	3.0	P	
SN250519008C046	3.09	1.2	3.0	P	
SN250519008C047	3.08	1.2	3.0	P	
SN250519008C048	3.08	1.2	3.0	P	
Supplementary information:					
- No fire or explosion					

7.3.8.1	TABLE: Vibration (Secondary batteries)					N/A
Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test (g)	Results	
Supplementary information:						

7.3.8.2	TABLE: Mechanical shock (Secondary batteries)					N/A
Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test (g)	Results	
Supplementary information:						

7.3.8B	TABLE: High rate charge (Secondary cells)				P
Sample No.	OCV at start of test, Vdc	Charging Current, A	Maximum Charging Voltage, Vdc	Results	
Samples charged at charging temperature upper limit (45°C)					
SN250519008C052	3.09	1.8	4.25	P	
SN250519008C053	3.08	1.8	4.25	P	
SN250519008C054	3.12	1.8	4.25	P	
SN250519008C055	3.11	1.8	4.25	P	
SN250519008C056	3.08	1.8	4.25	P	
Samples charged at charging temperature upper limit (-5°C)					
SN250519008C057	3.10	1.8	4.25	P	
SN250519008C058	3.11	1.8	4.25	P	
SN250519008C059	3.09	1.8	4.25	P	
SN250519008C060	3.09	1.8	4.25	P	
SN250519008C061	3.11	1.8	4.25	P	
Supplementary information:					
- No fire or explosion					

7.3.8D	Overcharge protection (Secondary batteries)				N/A
Sample No.	OCV at start of test, Vdc	Max. Voltage of cell/cellblock	Charging Voltage, Vdc	Results	
Supplementary information:					

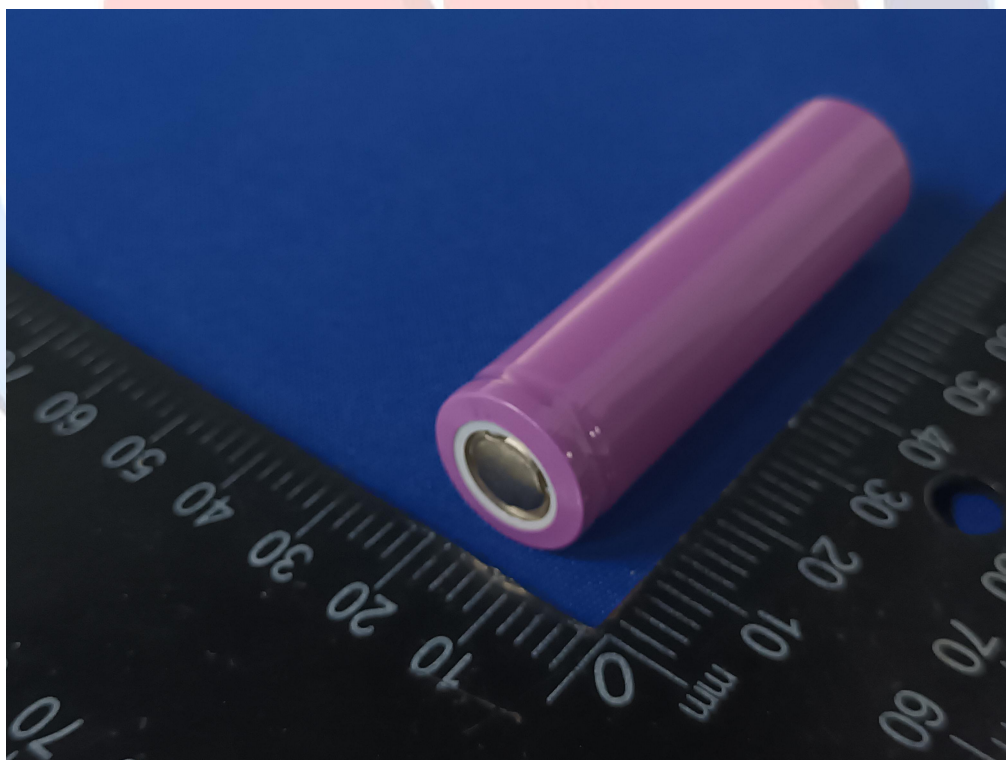
7.3.9	TABLE: Forced internal short circuit (Secondary cells)					P
Sample No.	Chamber ambient T (°C)	OCV before test (Vdc)	Particle location ¹⁾	Maximum applied pressure (N)	Results	
SN250519008C062	45	4.22	1	800	P	
SN250519008C063	45	4.21	1	800	P	
SN250519008C064	45	4.21	1	800	P	
SN250519008C065	45	4.21	1	800	P	
SN250519008C066	45	4.22	1	800	P	
SN250519008C06	45	4.22	1*	800	P	
SN250519008C068	45	4.21	1*	800	P	
SN250519008C069	45	4.21	1*	800	P	
SN250519008C070	45	4.22	1*	800	P	
SN250519008C071	45	4.21	1*	800	P	
SN250519008C072	-5	4.18	1	800	P	
SN250519008C073	-5	4.18	1	800	P	
SN250519008C074	-5	4.18	1	800	P	
SN250519008C075	-5	4.17	1	800	P	
SN250519008C076	-5	4.18	1	800	P	
SN250519008C077	-5	4.18	1*	800	P	
SN250519008C078	-5	4.17	1*	800	P	
SN250519008C079	-5	4.17	1*	800	P	
SN250519008C080	-5	4.18	1*	800	P	
SN250519008C081	-5	4.17	1*	800	P	
Supplementary information: ¹⁾ Identify one of the following: 1: Nickel particle inserted between positive and negative (active material) coated area. 2: Nickel particle inserted between positive aluminium foil and negative active material coated area. - No fire *Remark: No position 2 exist.						

D.2	TABLE: Internal AC resistance for coin cells				N/A
Sample no.	Ambient T (°C)	Store time (h)	Resistance Rac (Ω)	Results ¹⁾	
Supplementary information:					

Attachment 1: Photo documentation



Picture 1. Side view-1 of cell



Picture 2. Side view-2 of cell

Important

1. The test report is invalid if it is not affixed the official seal of the laboratory to it.
2. Copies of the test report without the official seal of the laboratory are invalid.
3. It is forbidden to copy the test report partially without the written approval of the laboratory.
4. The test report is invalid without the signatures of Approver, Reviewer and Testing engineer.
5. The test report is invalid if it is blotted out.
6. Objections to the test report must be submitted to CMC within 15 days.
7. The test report is valid for the tested samples only.
8. As for the Verdict, “-” means “no need for judgement”, “P” means “pass”, “F” means “fail” and “N/A” means “not applicable”.
9. This report is only responsible for the test results of the tested samples, and the test results and related judgment conclusions only reflect the evaluation of the tested samples. For the report and the content can not be used for commercial advertising, the use of direct or indirect losses and all legal consequences, the company does not assume any economic and legal responsibility.

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-- End of Report --